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William Lorenzo

Data Scientist / Data Scientist I

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Objective

Results-driven Data Scientist and AI Engineer with hands-on experience building machine learning models, geospatial data pipelines, and real-time simulation systems. Currently leading data and AI initiatives at Interlinked, developing wildfire prediction models using deep learning and physics-based methods. Passionate about applying machine learning, cloud infrastructure, and statistical modeling to solve high-impact real-world problems in healthcare and environmental systems.

Skills

Programming	Python, SQL, R, SAS, C++, Java
Machine Learning	TensorFlow, PyTorch, Nvidia CUDA, Scikit-Learn, Neural Networks (MLP, CNN, U-Net, ConvLSTM, GNN)
Data Science	Pandas, NumPy, Matplotlib, Seaborn, EDA, Feature Engineering
Cloud	AWS (S3, Lambda, CloudWatch), Terraform, Microsoft Azure, Data Pipelines
Databases	MySQL, SQLite
Data Visualization	Tableau, Matplotlib, Microsoft Excel
Other	Model Evaluation, Linear Algebra, Statistical Analysis, Optimization Techniques, One-Way/Two-Way ANOVA

Technical Experience

AI Engineer & Lead Data Analyst, Interlinked 2025 – Present
AI-powered wildfire prediction and disaster response platform

- Develop wildfire spread models combining physics-based methods (Rothermel, level-set) with ML surrogates.
- Implement deep learning architectures (U-Net, ConvLSTM, GNNs) for spatial-temporal wildfire forecasting.
- Build training datasets from satellite data (MODIS/VIIRS), fire perimeters, and historical incidents.
- Design AWS pipelines (S3, Lambda, CloudWatch) for scalable geospatial data ingestion and QA.
- Integrate real-time data using data assimilation techniques (e.g., Ensemble Kalman Filter).
- Lead data curation and support model validation, benchmarking, and deployment.

Projects 2025 – 2026

- **Myocardial Perfusion Segmentation with PyTorch U-Net** (ResearchGate, Mar 2026)
  Developed a U-Net model with a pretrained ResNet34 encoder for medical image segmentation (DICOM/NIfTI), achieving a Dice score of $\sim$0.86. Evaluated effects of data augmentation and optimization techniques, highlighting challenges with small datasets and class imbalance.
- **TensorFlow MLP for Heart Disease Prediction** (ResearchGate, Mar 2026)
  Built a neural network for binary classification using clinical data, applying preprocessing, feature scaling, and regularization techniques to improve model generalization and evaluation performance.
- **GPU Cloud Pricing Analysis (Lambda)** (ResearchGate, Dec 2025)
  Analyzed 300 GPU instances using statistical methods (regression, ANOVA, t-tests) to evaluate relationships between pricing, hardware performance, reliability, and customer segmentation.

Education

California State University - East Bay 2021 – 2026
Bachelor of Science in Statistics (Data Science Concentration)
GPA: 3.82

Honors & Awards

- **Magna Cum Laude, California State University - East Bay** Mar 2026
  Graduated with Latin honors in recognition of outstanding academic achievement; invited to the University Honors Convocation.
- **Dean's List, CSU East Bay** May 2024
  Recognized for high GPA and outstanding academic performance.
- **The National Society of Leadership and Success**
  Selected based on academic performance (GPA above 3.3).